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1. Background

In Singapore's current economic climate, polytechnic students face an increasing disparity between their fixed allowances and the rising costs of food, transport, and academic necessities. While the NTUC Income bursary for employees' children who are post-secondary students currently ranges from SGD 400 to SGD 550, there is growing concern that this benchmark is outdated. Despite the SGD 400 figure being a traditional baseline for monthly student expenses, current inflationary pressures necessitate an empirical review to determine if this amount remains a realistic reflection of today's cost of living.

This study utilises both descriptive and inferential analysis to provide a data-driven look at current student spending habits. The findings of this research are particularly significant for key stakeholders, including:

- **NTUC Income:** As a leader in social insurance, NTUC Income can leverage this data to refine their community support programs. Specifically, the data can justify the creation of a specialized education bursary of SGD 800 targeted at the children of NTUC Income employees and lower-income policyholders who are currently enrolled in polytechnics.

- **Targeted Support for Design & Media:** Our research specifically highlights students in the Design and Media faculty. These students face unique financial pressures, as their coursework often requires high-end hardware, specialized software subscriptions, and expensive physical materials for studio projects.

- **Strategic Allocation:** By analysing spending gaps, organizations like NTUC Income can move beyond "one-size-fits-all" support and implement a tiered bursary system—potentially doubling the standard aid to SGD 800 for those in high-cost vocational tracks to ensure they are not academically disadvantaged by rising costs.

Global Context: Rising student expenses are not unique to Singapore. In many countries, post-secondary students face increasing costs due to inflation, technology requirements, and higher living expenses. Understanding these trends helps contextualize our findings and highlights the importance of regularly reviewing student support policies to ensure they remain adequate and equitable.

By testing our primary survey data against the SGD 400 benchmark, we aim to provide these organizations with the statistical modelling necessary to make informed decisions regarding student welfare, targeted financial aid, and the sustainable support of Singapore's future workforce.

2. Research Statements

One Sample T-Test:
Polytechnic students' average monthly expenditure is higher than the average student budget of SGD 400.

Paired T-Test:
Design Media students spend more per month compared to students in other courses.

3. Hypothesis

Test Type	Research Statement	Null	Alternate
One-sample t-test	Polytechnic students' average monthly expenditure is higher than the average student budget of SGD 400.	H0: μ <= 400	H1: μ > 400
Paired t-test	Design & Media students spend more per month compared to students in other courses.	H0: μ1 <= μ2	H1: μ1 > μ2

4. Data Collection

Secondary Data (for one sample T-Test)

- “Parents give their child an average of $14 a day”
- [How much pocket money do Singapore children get? It's $4 to $14 a day | The Straits Times]

Primary Data (Survey)

Google Forms

- Target Population : Students in Polytechnic
- Methods of Data Collection: Google Forms
- Sample Size: n = 21 (As this is a preliminary study)
- Variables collected:
    - How much do you spend per month?
    - Which field of study are you from?
    - Where do you get your monthly allowance from?
    - Which category represents the largest portion of your monthly expenditure?
    - Which factor most influences your spending decisions?

SRM Data

	How much do you spend per month?	Which field of study are you from?	Where do you get your monthly allowance from?	Which category represents the largest portion of your monthly expenditure?	Which factor most influences your spending decisions?
2	$300.00	Applied Science	Parents / Guardian, Work, Savings	Food and Dining	Necessity / Essential needs
3	$500.00	Applied Science	Parents / Guardian, Work, Savings	Entertainment / Leisure	Financial planning / Budgeting
4	$200.00	Business	Parents / Guardian	Food and Dining	Emotional factors / Impulse buying
5	$450.00	Business	Parents / Guardian, Work	Shopping (Clothes / Gadgets)	Peer influence / Social trends
6	$1,000.00	Design & Media	Parents / Guardian, Work	Shopping (Clothes / Gadgets)	Emotional factors / Impulse buying
7	$300.00	Design & Media	Parents / Guardian, Work	Food and Dining	Sales / Discounts / Promotions
8	$600.00	Design & Media	Parents / Guardian, Work, Savings	School Supplies / Fees	Necessity / Essential needs
9	$200.00	Design & Media	Parents / Guardian	Food and Dining	Necessity / Essential needs
10	$6,700.00	Design & Media	Catfishing	Shopping (Clothes / Gadgets)	Sales / Discounts / Promotions
11	$650.00	Design & Media	Parents / Guardian, Work, Savings	School Supplies / Fees	Necessity / Essential needs
12	$550.00	Design & Media	Parents / Guardian	School Supplies / Fees	Necessity / Essential needs
13	$670.00	Design & Media	Parents / Guardian	School Supplies / Fees	Necessity / Essential needs
14	$720.00	Design & Media	Parents / Guardian	School Supplies / Fees	Necessity / Essential needs
15	$630.00	Design & Media	Work	School Supplies / Fees	Necessity / Essential needs
16	$490.00	Design & Media	Parents / Guardian, Scholarships / Awards	Shopping (Clothes / Gadgets)	Peer influence / Social trends
17	$300.00	Engineering	Parents / Guardian	Food and Dining	Necessity / Essential needs
18	$200.00	Engineering	Parents / Guardian	Food and Dining	Peer influence / Social trends
19	$250.00	Health & Social Sciences	Parents / Guardian, Work, Savings	Food and Dining	Sales / Discounts / Promotions
20	$360.00	Health & Social Sciences	Parents / Guardian	Food and Dining	Peer influence / Social trends
21	$500.00	Information Technology	Parents / Guardian, Work	Transportation	Financial planning / Budgeting
22	$400.00	Information Technology	Parents / Guardian, Work, Savings	Food and Dining	Necessity / Essential needs

5. Hypothesis Testing Plan

To investigate student spending habits, our group implemented the following statistical plan:

- Significance Level: We have set the level of significance at a = 0.05$. This means we are willing to accept a 5% risk of committing a Type I error (concluding there is a difference when one does not exist)

- One-Sample T-Test: * Direction: One-tailed (Right-tailed), as we are testing if the mean spending is specifically *higher* than the benchmark.
    - Benchmark: μ0 = 400 SGD.
    - Degrees of Freedom: df = n - 1 = 20.

- Paired T-Test (Group Comparison): * Note: Although this test compares two different groups (Design & Media vs. Others) as approved by our tutor, we are treating this as a comparison of means to identify if specific academic disciplines face higher financial pressures.
    - Direction: One-tailed (Right-tailed), testing if Design & Media students spend *more* than others.
    - Variables: The independent variable is the "Field of Study," and the dependent variable is "Monthly Expenditure."

- Rejection Rule: We will reject the Null Hypothesis (H0) if the calculated p-value < 0.05.

- Software: Calculations were performed using Python (SciPy library) for accuracy and data visualization.

6. Results

Summary of Sample Data

The following table summarizes the descriptive statistics for our survey respondents (n = 21):

Metric	All Students	Design & Media Students	Other Courses
Sample Size (n)	21	11	10
Mean Expenditure	$760.48	$1,137.27	$346.00
Standard Deviation (s)	$1,376.21	$1,856.93	$114.23

Statistical Test Outcomes

Test 1: One-Sample T-Test (vs. $400 Benchmark)

- Test Statistic (t): 1.2003
- p-value: 0.1220
- Decision: Fail to Reject H0 (since p > 0.05).
- Interpretation: While the sample mean ($760.48) is higher than the $400 benchmark, the high variability in spending among students means we do not have enough statistical evidence to claim that the average polytechnic student spends significantly more than $400 at a 95% confidence level.

Test 2: Paired T-Test (Design & Media vs. Others)

- Test Statistic (t): 1.3420
- p-value: 0.0977
- Decision: Fail to Reject H0 (since p > 0.05).
- Interpretation: On average, Design & Media students in our sample spend $1,137.27 compared to $346.00 for other students. However, due to the small sample size and large variance within the Design & Media group, the difference is not statistically significant at the 5% level.

7. Discussion and Conclusion

Summary of Findings

Our research set out to investigate whether the traditional SGD 400 monthly budget is still a realistic benchmark for polytechnic students in Singapore's current inflationary economy. Our descriptive analysis of the primary data revealed a sample mean expenditure of SGD 760.48, which is significantly higher than the benchmark. Furthermore, our comparison of academic tracks showed that Design and Media students face the highest financial burden, with an average monthly spend of SGD 1,137.27.

Interpretation and Implications:

Although our one-sample t-test ($p = 0.1220$) and independent t-test ($p = 0.0977$) did not reach the 5% significance level—largely due to high variance and a small sample size—the raw data points toward a clear "spending gap".

- The SGD 400 Benchmark: This figure, often used by parents and institutions to set allowance or bursary levels, appears outdated given that 19 out of 21 surveyed students reported spending above this amount once extreme outliers were considered.
- Faculty-Specific Costs: The high mean for Design and Media students supports our initial background assertion that these students require additional funds for specialized school supplies, such as high-spec laptops and physical materials.

Recommendations for Stakeholders (NTUC Income)

Based on the insights gained from this study, we provide the following recommendations for NTUC Income regarding their bursaries for employees' children:

1. **Adjust the Base Bursary Level:** NTUC Income should consider raising the baseline bursary from the current **SGD 400–550** range to a minimum of **SGD 600**. This adjustment would better reflect the rising costs of food and transport identified in our survey.

2. **Implement a Tiered "Specialized Track" Bursary:** We recommend a specific **SGD 800** bursary for students enrolled in resource-heavy courses like Design and Media. This provides targeted support for the unique "school supply" costs associated with these diplomas, ensuring these students are not at a financial disadvantage compared to their peers in other courses.

Final Conclusion

In conclusion, while a **SGD 400** budget may have been sufficient in the past, real-world data suggests that polytechnic students—particularly those in creative fields, like the school of Design and Media—require more substantial support. By adopting a tiered bursary model of **SGD 800**, stakeholders like NTUC Income can more effectively support the next generation of the Singaporean workforce in navigating an increasingly expensive economic landscape.

Member Contributions:

1. **Stephanie**
   - Researched and Identified data sources
   - Conducted the one-sample t-test analysis
   - Formulated the hypotheses for the one-sample t-test
   - Drafted the results, outcome, discussion and conclusion sections based on the one-sample t-test

2. **Yunshan**
   - Wrote the background and context of the study
   - Formulated the hypotheses for both the one-sample and paired t-tests
   - Designed the Google Form survey questions
   - Cleaned and prepared the dataset for analysis
   - Drafted the discussion and conclusion sections for the paired t-test.

3. **Julia**
   - Designed the Google Form survey questions
   - Cleaned and validated Google Form survey data
   - Conducted the paired t-test analysis
   - Drafted the discussion and conclusion sections relating to the paired t-test

4. **Kahlen**
   - Cleaned and validated Google Form survey data
   - Conducted the one-sample t-test analysis
   - Researched and Identified data sources
   - Drafted the results, discussion, and conclusion sections based on the one-sample t-test

5. **Rainie**
   - Cleaned and validated Google Form survey data
   - Conducted the paired t-test analysis
   - Design research statement
   - Drafted the discussion and conclusion sections for the paired t-test.